

SMART INDIA HACKATHON 2026



EcoStay IoT: Intelligent Edge-Driven Energy Optimizer for Budget Hotels

Problem Statement ID – 26221

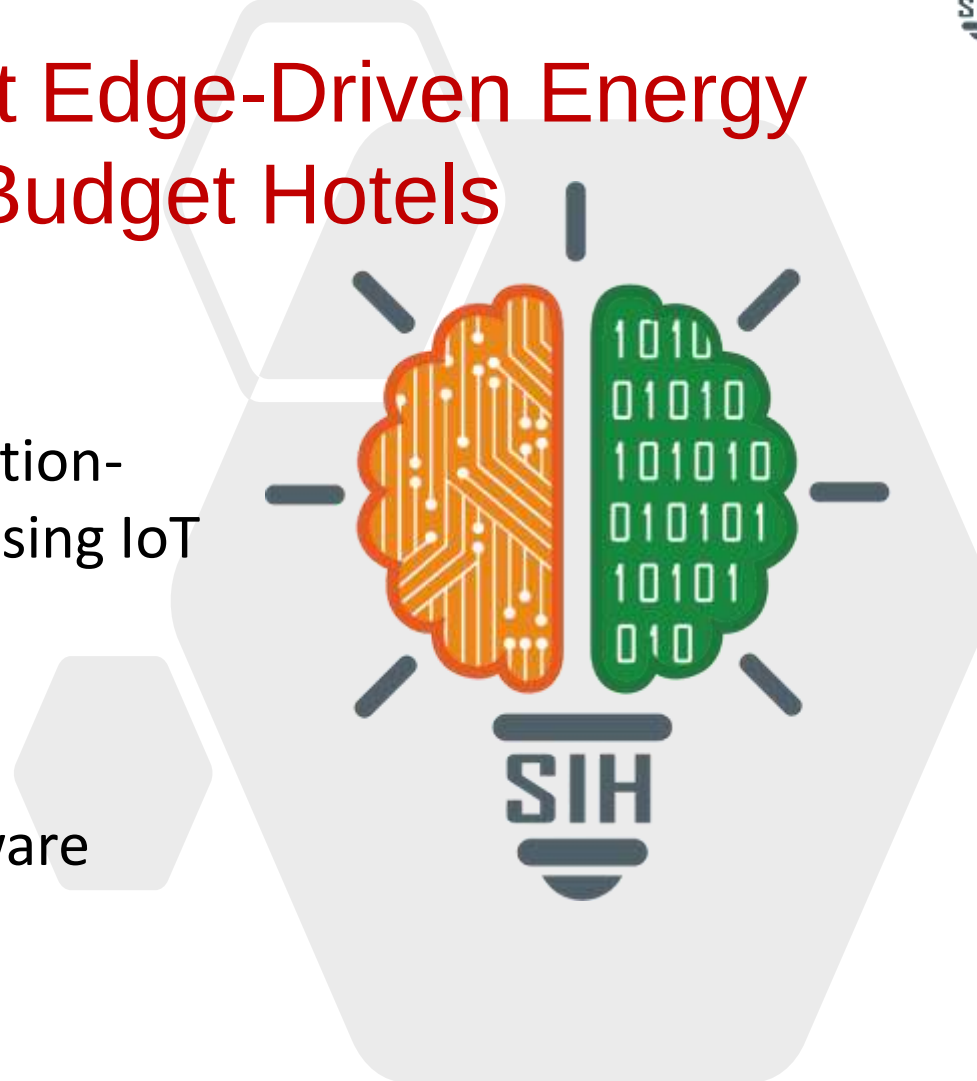
Problem Statement Title – Student Innovation-
Energy Optimization in Hotels using IoT

Theme – Travel & Tourism

PS Category- Software/Hardware – Hardware

Team ID – Pacathon 2.0

Team Name (Registered on portal) – ECE NAVIGATORS



EcoStay IoT: Intelligent Energy Optimizer for Budget Hotels

Detailed explanation of the proposed solution

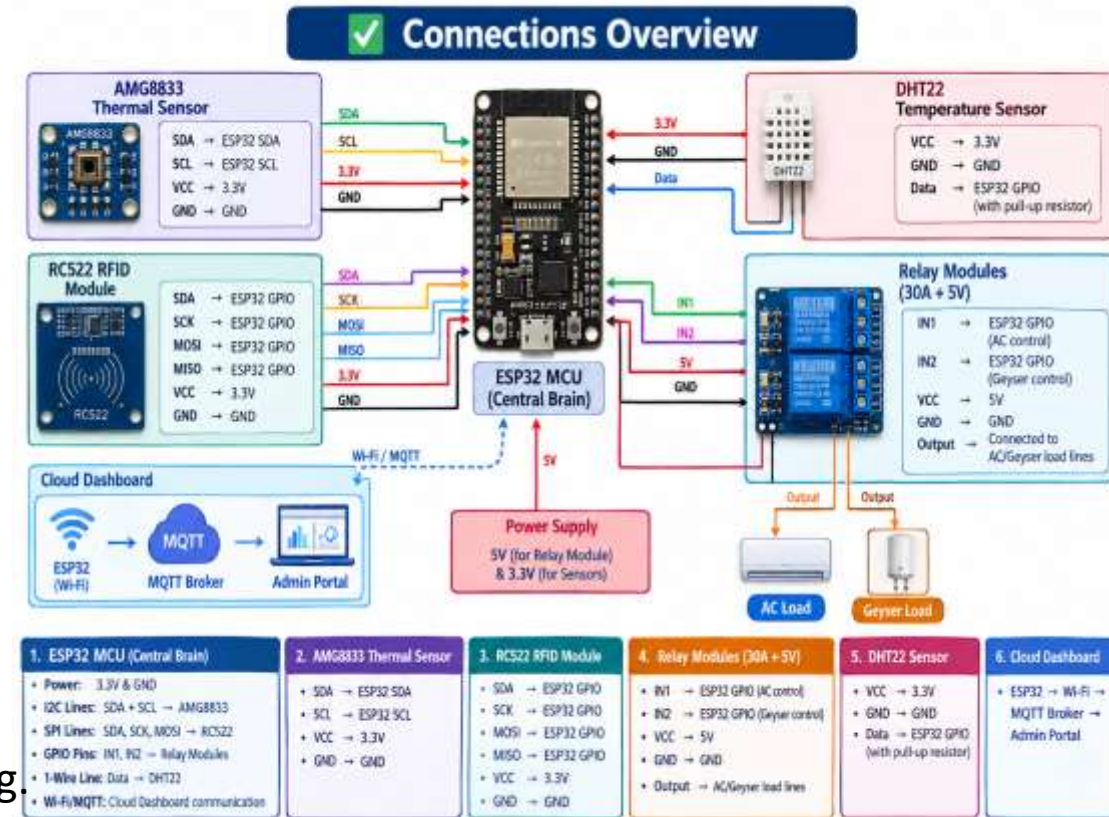
- **ESP32 microcontroller** acts as the central brain.
- **AMG8833 thermal sensor** tracks room occupants.
- **30A isolated relays** cut power to heavy appliances.
- Automatically cuts AC/Geyser power when room is empty.

How it addresses the problem

- Eliminates heavy power waste in empty hotel rooms.
- Cuts hospitality electricity bills by up to 35%.
- Overrides fake keycards using live heat tracking.
- Sends real-time data to a central management dashboard.

Innovation and uniqueness of the solution

- Tracks body heat, not motion—never cuts power while sleeping.
- **Affordable (~₹1,600 per room)** vs ₹12,000–₹60,000 commercial systems.
- **100% privacy-safe** array—uses zero optical lenses or cameras.
- Low cost and **plugs directly behind existing switchboards**.
- **Cannot be tricked** with spare cards unlike traditional hotel slots.

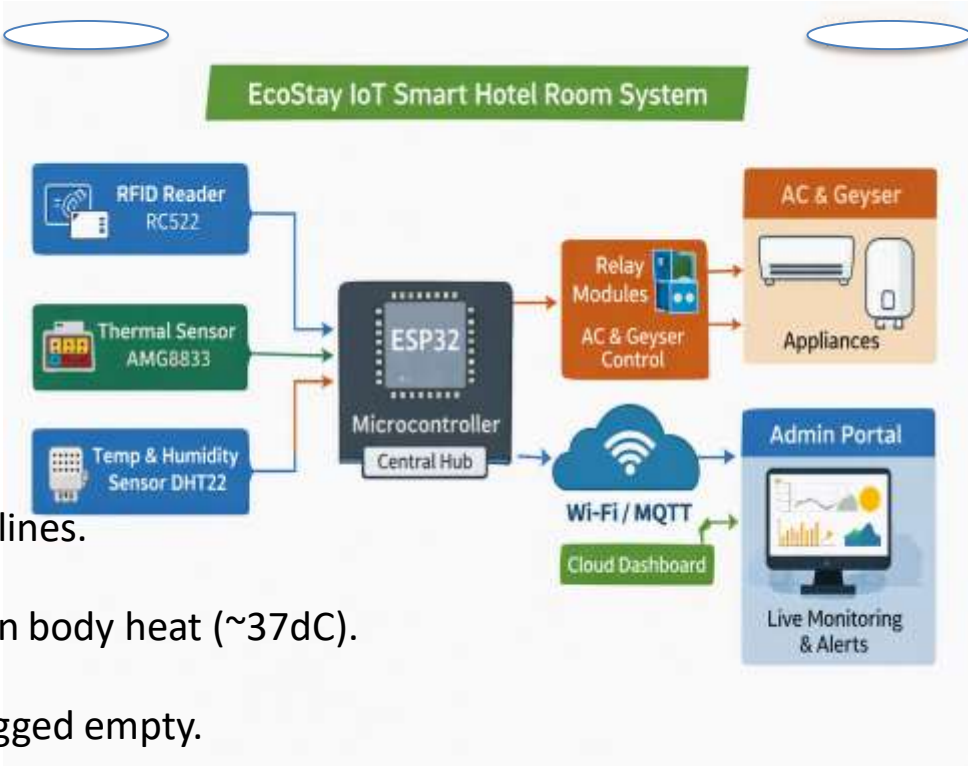


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Technologies to be used:

 - Core Board: [ESP32 MCU](#) with built-in Wi-Fi and Bluetooth.
 - Main Sensor: [AMG8833 Thermal Grid](#) using simple I2C lines.
 - Door Access: **RC522 RFID Node** using SPI communication.
 - Power Control: **30A Isolated Relays** to switch heavy AC lines.
 - Software Stack: **C++ (Arduino IDE)** and lightweight **MQTT protocols**.

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Methodology and process for implementation:

 - RFID Swipe:** Guest entry triggers the RFID card node to activate main power lines.
 - Thermal Scan:** The AMG8833 grid continuously monitors the room for human body heat (~37dC).
 - Timeout Check:** If the room registers 0 occupants for over 5 minutes, it is flagged empty.
 - Load Isolation:** The ESP32 switches high-current relays to cut power to AC and geysers.
 - Dashboard Sync:** Live room status and energy savings data are uploaded over Wi-Fi.



🔧 Analysis of the feasibility of the idea:

- ❑ **Low Component Cost:** Total hardware package costs ~₹1,640, making it highly affordable.
- ❑ **Retrofit Design:** No wall breaking needed; fits directly inside existing modular hotel switchboards.
- ❑ **Rapid Return on Investment:** Saves enough power to pay for itself completely within **3 to 4 months**.

🔧 Potential challenges and risks:

- **High Ambient Heat:** High Indian summer temperatures can mask human thermal signatures.
- **Blind Spots:** Bathroom areas or deep room corners are out of the sensor's direct view.
- **Relay Sparking:** Heavy AC inductive loads can burn out cheap mechanical relay switches over time.

🔧 Strategies for overcoming these challenges:

- ✓ **Dynamic Calibration:** Use a **DHT22 sensor** to auto-adjust thermal array baselines based on room temperature.
- ✓ **Smart Placement:** Mount the node directly on the ceiling over the bed/desk area to target primary occupancy zones.
- ✓ **Solid-State Upgrades:** Upgrade to **Solid-State Relays (SSRs)** with zero-cross detection to stop electrical arcing.



📱 Potential impact on the target audience:

- **For Hotel Owners:** Drastically drops utility overhead costs, boosting net profit margins.
- **For Hotel Managers:** Provides automated, room-by-room power tracking dashboards.
- **For Tourists:** Offers automated, hassle-free room environments without false power cut-offs.
- ROI in 2–3 months for 10 rooms.
- Hotels save ~35% electricity.

📱 Benefits of the solution:

- **Economic Benefit:** Trims electricity wastage by **35%**, recovering full equipment costs in 3 to 4 months.
- **Environmental Benefit:** Lowers carbon footprints across tourist hubs by cutting grid electricity wastage.
- **Social Benefit (Privacy):** Tracks room occupants using basic temperature grids, keeping user privacy **100% safe from optical cameras**.



RESEARCH AND REFERENCES



Details / Links of the reference and research work:

Thermal Matrix Research: Looked into IEEE papers tracking motionless indoor occupants with low-resolution infrared arrays.

ESP32 Core Documents: Followed official Espressif datasheets to build stable, dual-core Wi-Fi/MQTT sensor mesh systems.

AMG8833 Sensor Guide: Followed Panasonic documentation for calibrating 8x8 thermopile matrix sensitivity levels.

Hotel Energy Studies: Analyzed Bureau of Energy Efficiency (BEE) India reports highlighting HVAC losses in commercial spaces.

RFID Safety Protocols: Integrated standard cryptographic verification logic based on ISO/IEC 14443A entry tracking frameworks.

- Research on hotel energy wastage (~35%).
- Existing BMS solutions costing ₹12,000–₹60,000 per room.
- IoT sensor datasheets (ESP32, AMG8833, RC522, DHT22).
- MQTT protocol documentation.
- SIH problem statement reference.

THANK YOU